

Question ID: 7d5c32e6

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Expression of Ideas	Rhetorical Synthesis	Hard

Question

While researching a topic, a student has taken the following notes:

- The fifth Solvay Conference on Physics was held in 1927.
- It brought together twenty-nine of the era’s preeminent scientists to discuss the emerging field of quantum theory.
- The conference famously featured a debate between physicists Albert Einstein and Niels Bohr.
- Bohr proposed that subatomic entities like electrons had only probable realities until they were observed.
- Einstein argued that subatomic entities like electrons had a reality independent of observation.
- Bohr’s position, later called the Copenhagen interpretation, remains the most widely accepted theory of quantum mechanics.

The student wants to place Einstein’s argument within its historical context. Which choice most effectively uses relevant information from the notes to accomplish this goal?

Answer

- A. During the dawn of quantum theory, Einstein maintained the independent reality of some subatomic entities, although Bohr’s opposing interpretation would become the widely accepted view.
- B. At the 1927 Solvay Conference on Physics, Einstein disagreed with Bohr’s argument that subatomic entities like electrons had a reality independent of observation.
- C. The attendees of the 1927 Solvay Conference were among the preeminent scientists of their era, including Einstein, who opposed Bohr’s proposal.
- D. In 1927, Einstein and Bohr engaged in a famous debate; Bohr’s argument, later called the Copenhagen interpretation, would remain popular decades after.

Correct Answer: A

Rationale

Choice A is the best answer. The sentence places Einstein’s argument within the historical context of the development of quantum theory, noting that his argument—made during the dawn of the field—conflicted with Bohr’s argument, which became the widely accepted view.

Choice B is incorrect. The sentence misrepresents information from the notes, attributing the argument that electrons had a reality independent of observation to Bohr, not Einstein. In addition, while the sentence provides the date of the conference, it doesn’t place Einstein’s argument in the context of the development of quantum theory. Choice C is incorrect. The sentence indicates that Einstein attended the 1927 Solvay Conference; it doesn’t identify Einstein’s argument or place it in the historical context of the development of quantum theory. Choice D is incorrect. The sentence explains that Einstein and Bohr had a famous debate in 1927 and that Bohr’s argument remained popular decades afterward; it doesn’t identify Einstein’s argument or place it in the context of the development of quantum theory.